

17 THE INTERNET

The Internet, short for International Network, is a worldwide collection of interconnected networks and the interconnection of computers, people and information which forms the global communications system. The Internet is also known as cyberspace. It offers many services with different facilities.



To connect to the Internet you need a broadband connection for fast access. The internet access can be fixed or mobile, e.g. xDSL, fibre optic cable, wireless connection (Wi-Fi, cellular). Next, you need an account

with a company called Internet Service Provider (ISP) which provides Internet connections and gives you access to the Internet for a monthly fee (flat rate) or you sign up to a mobile phone operator's services. The quality and speed of your access is determined by several factors, e.g. connectivity, the type of connection, interference, the capacity of your device, the quality of signal, etc.

Internet address, TCP/IP model

The Internet uses internet (network) addresses. Every computer (node) in the network is given a unique IP address (Internet Protocol address) which identifies a computer on the Internet. Internet addresses are numeric, a 32-bit code number (IPv4) or a 128-bit code (IPv6). Since they are difficult for people to remember, the Internet also uses domain names which were developed along with the Domain Name System (see Chapter 18 The World Wide Web).

The Internet uses protocol suite known as TCP/IP (Transmission Control Protocol/Internet Protocol). It is the set of standards which enable computers to communicate, transmit and deliver data packets and to interconnect different types of networks (local area, public data networks, and wide area networks) based on packet switching.

TCP/IP model is divided into four layers:

4	Application Layer	It is the top-most layer which is related to layers 7-5 of the OSI model. It provides communication functions.	Email protocols: SMTP, POP3, IMAP
			Web: HTTP, HTTPS, DNS
			File sharing: FTP, SMB, NFS
3	Transport Layer	It provides message segmentation and transportation.	TCP, UDP
2	Internet Layer	It enables logical addressing to deliver data packets, traffic routing and traffic control.	IPv4, IPv6, ICMP, ICMPv6
1	Network Access Layer (also Network Interface Layer)	It combines the physical and data link layer of the OSI model.	Ethernet, ARP, Wi-Fi

Overview of the most used Internet services

The Internet provides many services for information sharing, communication, entertainment and various other activities. It helps people with work, studies and research. In fact, this virtual world has changed the way people communicate, interact and work. It has also transformed numbers of jobs.

1) The World Wide Web is an information service on the Internet that allows document pages to be accessed. It is a system of interlinked hypertext documents (files and pages containing links to other documents called hyperlinks). Commonly it uses HTTP and HTTPS protocols.

2) Electronic mail or email for sending and receiving messages and attached files (email attachments).

3) File sharing services, e.g. File Transfer Protocol (FTP) is a standard for transferring files between a client and a server. Server Message Block (SMB) provides shared access to files, printers and serial ports, and communications between nodes on a network. Network File System (NFS) is a file system protocol that allows a user access files over a network on a remote computer, often used with Unix operating systems.

4) Communication and social media allow a new type of social interaction. Examples:

- *Chats* (chat rooms) allow a conversation on the Internet where the users can communicate in real time.
- *Online discussion forums* (Internet forums) are series of posts, usually on a related topic.
- *Instant Messaging (IM)* is a type of communication among users which allows exchanging text messages in real time between two or more users logged into IM services, e.g., Facebook Messenger, WhatsApp Messenger, etc.
- *Social networking sites* allow users to share messages, upload photos, videos, to post opinions, e.g. Facebook, Google+, Twitter, Instagram, LinkedIn, YouTube, etc.

- *Blogging* – writing blogs by bloggers or blog editors (video blogs include videos).
- *Internet telephony* allows calling over a data network, e.g. VoIP (Voice over Internet Protocol) technology enables the users to make phone calls that can be sent via Internet.
- *Video conferencing* (virtual meetings over long distances) is a form of communication over a network that uses video cameras and display screens so that people can talk to and see each other.

5) Entertainment:

- Internet TV (IPTV),
- Internet radio,
- Streaming media (the real-time delivery of digital media),
- Webcast (live broadcast over the Internet),
- Online games, virtual reality.

6) E-services:

- *E-Government* allows citizens to have access to public services.
- *E-Commerce* includes commercial activities and doing business on the Internet.
- *Online shopping* enables people to sell and buy goods in an e-shop.
- *Online banking* allows instant access to your account, making payments and managing your finances through the Internet.
- *E-Health* refers to healthcare services and information delivered through the Internet.
- *E-learning* provides learning management systems to create online learning courses, e.g. Moodle, Microsoft Teams, Google Classroom, etc.

7) The Internet of Things (IoT) is the interconnection of technical devices, vehicles and other objects with embedded technology to the Internet, such as cars, set-top boxes, TV sets, electrical appliances, etc. This concept aims to connect people, data and things in the future.

Note:

The WWW, email and e-services are discussed in more detail in the following chapters.



Social networking sites

VOCABULARY

account /ə'kaʊnt/	účet
application layer /,æp.li'keɪ.jən ,leɪ.ər/	aplikační vrstva
chat /tʃæt/	konverzace po síti
cyberspace /'saɪ.bə.speɪs/	kybernetický prostor
cyberstalking /'saɪ.bə.stɔː.kɪŋ/	kybernetické obtěžování
e-Commerce /,iː'kɒm.ɜːs/	obchodování na internetu
email (electronic mail) /'iː.meɪl/	elektronická pošta

File Transfer Protocol /faɪl ˈtrænsˈfɜːr ˈprəʊ.tə.kəl/	protokol pro přenos souborů
flat rate /flæt ˈreɪt/	paušální poplatek
instant messaging /ˌɪnstənt ˈmesɪdʒɪŋ/	rychlé zasilání zpráv
interconnect /ˌɪn.tə.kəˈnekt/	propojit
internet layer /ˌɪn.tə.net ˌleɪ.ər/	síťová vrstva
Internet of Things /ˌɪn.tə.net əv ˈθɪŋz/	internet věcí
Internet Protocol /ˌɪn.tə.net ˈprəʊ.tə.kəl/	protokol síťové vrstvy
Internet Service Provider /ˌɪn.tə.net ˈsɜː.vɪs prəˈvaɪ.dər/	poskytovatel internetového připojení
Internet telephony /ˌɪn.tə.net təˈlef.ə.ni/	internetová telefonie
IP address /aɪˈpiː əˈdres/	internetová adresa
learning management system /ˌlɜːnɪŋ ˈmænɪdʒmənt sɪstəm/	systém pro řízení výuky
messenger /ˈmes.ɪn.dʒər/	internetový komunikátor
network access (interface) layer /ˌnet.wɜːk ˈæk.ses ˌleɪ.ər/	vrstva síťového rozhraní
packet switching /ˈpækɪt ˌswɪtʃɪŋ/	přepojování paketů
social network /ˌsəʊ.ʃəl ˈnet.wɜːk/	sociální síť
TCP/IP protocol suite /tiː siː ˌpiː ˈpiː / ˈprəʊ.tə.kəl swɪt/	rodina protokolů TCP/IP
Transmission Control Protocol /trænzˌmɪʃ.ən kənˌtrəʊl ˈprəʊ.tə.kəl/	protokol řízení přenosu
transport layer /ˈtræn.spɔːt ˌleɪ.ər/	transportní vrstva
VoIP /vɔɪp/	hlasové přenosy po internetu
worldwide, world-wide /ˌwɜːldˈwaɪd/	celosvětový

NOTES

QUESTIONS

- 1 What is the Internet?
- 2 How can we connect to the Internet?
- 3 What is ISP?
- 4 Explain 4-layer TCP/IP model (purpose, layer names, protocol suite).
- 5 Can social media affect people negatively?
- 6 What are the advantages of social media (social networking)?
- 7 What type of communications do you prefer?
- 8 What do you use the Internet for? Which facilities available on the Internet do you use?

ACTIVITY

(Web-based)

1. Learn about network and information security and visit the ENISA website:
<https://www.enisa.europa.eu/>
2. Find out more about cyber education at:
<https://www.enisa.europa.eu/topics/cybersecurity-education>
3. Take the cybersecurity quiz at:
<https://www.enisa.europa.eu/topics/cybersecurity-education/nis-in-education>